A Comparative Analysis of Deep Learning Models for Fruit Ripeness Classification Using CNN and Transformer Architectures

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Abstract: Ripeness detection of fruits is one of the vital tasks in farming, logistics management, and food quality assurance due to the impact it has on storage conditions, prices, and customer satisfaction. Traditionally, ripe fruits have been classified manually by observing physical features like color and texture, which are quite subjective and inefficient. In order to improve the process of fruit ripeness classification, this research will implement deep learning models such as CNNs and transformers. For this study, there will be a total of nine different models that will be compared, namely the three customized Convolutional Neural Networks architecture, AlexNet, DenseNet201, InceptionResNetV2, and the Vision Transformer (ViTB/32), DeiT-Small, Swin Transformer Tiny as transformer models. The dataset is preprocessed by resizing and normalizing the images to have the same input format. The results from the experiment have indicated that the DenseNet201 algorithm has achieved the highest accuracy rate of 94.10% followed by the InceptionResNetV2 architecture which has an accuracy rate of 91.25%. This is while the custom CNN architectures have performed competitively well while using far fewer parameters. On the other hand, the transformer-based architectures have been found to perform poorly, where the ViT-B/32 model obtained an accuracy rate of 75.38%.

Keywords: Fruit Ripeness Classification, Deep Learning, Convolutional Neural Networks (CNN), Vision Transformer (ViT), DenseNet201, InceptionResNetV2, Image Classification, Computer Vision, Agriculture Technology, Model Comparison.

Introduction:

Fruit ripeness detection is an essential part of modern-day agriculture, processing of food, and distribution systems. It is very important to establish the correct stage of maturity as this will help in setting the right time for harvesting. The common method that is used by people to detect the stage of maturity of fruits is through manual inspection where several parameters are considered. These parameters include color, size, and texture. Manual inspection, despite being

widely used, is highly subjective. Therefore, there exists a need for automated fruit ripeness detection techniques[1]. Deep learning algorithms have been developed owing to the advancement in artificial intelligence and computer vision techniques that aid in the automation of image-classification tasks. The main algorithms for image analysis in the modern world are convolutional neural networks, because they enable automated learning of hierarchical features extracted directly from images [11]. CNNs capture spatial features like edges, textures, and colors, and therefore perform exceptionally well when it comes to classifying objects that share visual similarity, for example, different ripeness levels of fruits [8]. Some examples of CNN architectures include AlexNet, ResNet, and DenseNet [10][5][7].

CNNs have proven themselves to be highly effective when solving various computer vision problems like object detection, biomedical image processing, and agriculture monitoring [8], [14]. For example, DenseNet incorporates densely connected layers, thus ensuring efficient feature reutilization and gradient backpropagation and improving model performance as well as vanishing gradient prevention [7]. Another successful approach to solving the degradation problem is provided by residual neural networks that enable learning in very deep networks thanks to skip connections [5].

Recently, computer vision architecture development has seen another promising trend with the introduction of transformer-based designs. The initial idea behind transformers was to utilize self-attention mechanisms in modeling long-range relationships between elements in sequences used in natural language processing [24]. Following this logic, Vision Transformer (ViT) applies the technique to solving image classification tasks by segmenting images into patches and treating these segments as sequences [3]. Data-efficient versions of ViT such as DeiT and Swin Transformer were then proposed [12], [24].

Although transformers have promising capabilities, transformer models typically need extensive datasets to perform efficiently because of the absence of spatial biases [3]. CNNs, on the other hand, take advantage of locality, and hence they become preferable in applications that work with small datasets [9]. This makes the decision between CNN and transformers a relevant area of study in scenarios where data scarcity prevails.There exist a variety of works that apply deep learning to classify images in agriculture as well as detect the ripeness of fruits [1], [8], [23]. Yet most of these papers investigate the performance of one deep learning architecture only, which makes the investigation into the performance of diverse architectures

in identical conditions impossible. Moreover, very few studies compare CNN and transformer models for classifying the ripeness of fruits.

In light of the above, this paper introduces a comparative study involving several deep learning models for classifying fruits based on ripeness levels. For that purpose, the FreshCheck tomato data set was selected. It contains a series of real-life photographs divided into unripe, ripe, and overripe categories. In total, nine deep learning models were tested, namely, custom-designed convolutional neural networks (CNNs), AlexNet, DenseNet201, InceptionResNetV2, and transformers, viz., ViT-B/32, DeiT-Small, and Swin Transformer Tiny. Training and preprocessing procedures remained consistent across all experiments.

First, this study aims to find the best-performing model among those tested. Secondly, the authors are interested in the difference in performance between the two types of deep learning architectures: CNNs and transformers. The former type includes custom-designed models, AlexNet, and DenseNet201, while the latter covers ViT-B/32, DeiT-Small, and Swin Transformer Tiny. As the metrics for measuring model performance, accuracy, precision, recall, and F1-score were used.

The contributions made by the present paper are three-fold. Firstly, it makes a comparative assessment of CNN and transformer models' performance in the domain of fruit ripeness classification. Secondly, the connection between model complexity and performance is revealed. Finally, some limitations associated with transformers in small data sets are discussed.

Literature Reviews: The technique of deep learning has become the most successful paradigm in the field of computer vision due to its ability to considerably outperform conventional image processing algorithms in many tasks of object recognition. Conventional machine learning algorithms were built on the basis of artificial feature creation, such as color histograms, shape descriptions, and textures. However, in this case, it was difficult to ensure good performance on datasets due to varying illumination conditions, complex backgrounds, and orientation problems [11], [4]. The creation of deep neural networks enabled systems to create their own features and classify objects without human intervention, resulting in an improvement in classification accuracy [11], [4]. It is worth mentioning that the use of AlexNet was a revolution in the deep learning community, since deep CNNs showed superior results in comparison with classical algorithms in image classification tasks [10].

In the wake of the success of AlexNet, the emphasis shifted towards constructing deeper architectures for better results. The work by Simonyan and Zisserman introduced VGG networks, showcasing that depth was an important factor when constructing networks, and deeper networks yielded better results with the help of small convolutional filters [21]. As models increased in their depths, challenges in their construction became more apparent, such as issues like vanishing gradient problems and degradation of results during training. To solve this problem, He et al. proposed Residual Networks (ResNet) architecture in which the inclusion of skip connection allowed the propagation of gradients through the network more effectively during training [5]. As a result of this innovation, very deep networks could be constructed and trained successfully. Following suit, DenseNet architecture was introduced in which each layer uses the feature map produced by all the previous layers for its training [7].

Deep learning algorithms have been widely used in agriculture applications owing to their capability of dealing with complicated visual data from the real world. Kamilaris and Prenafeta-Boldú presented a systematic review about the applications of deep learning in agriculture, pointing out its outstanding performance in crop classification, plant disease detection, and yield estimation [8]. This research indicated that deep learning algorithms based on CNNs perform better than conventional techniques in cases with complex background environments and higher variability. In addition, Mohanty et al. showed that deep CNN models were able to classify plant diseases based on large amounts of datasets with high accuracy and reliability [14].

In particular, multiple research papers have been devoted to fruit quality evaluation and fruit ripeness classification based on deep learning algorithms. For instance, Aherwadi et al. evaluated the possibility of predicting the fruit maturity stage with the help of convolutional neural networks and obtained satisfactory results concerning recognizing various ripeness stages [1]. It should be noted that CNN-based models are able to detect very subtle visual changes (changes in color or texture). Tapia-Mendez et al. analyzed deep learning algorithms for fruit classification problems and emphasized the potential of convolutional neural networks for solving ripeness-related problems [23].

Despite the high performance of CNN-based models, it should be noted that another important family of algorithms – namely, transformer-based models – have emerged relatively recently. For example, Vaswani et al. proposed a transformer framework that is based on self-attention to consider the relationships between all elements of the sequence [24]. Then, the transformer

idea was adopted for image processing purposes; in particular, Dosovitskiy et al. proposed Vision Transformer (ViT), which considers patches in images as sequential input [3]. Thus, transformers differ from CNNs because the latter models focus on spatial information whereas transformers focus on the relationships between elements of an image.

In another study, Liu et al. presented the Swin Transformer architecture that utilizes the concept of hierarchical structure and shifted windows to increase the efficiency and scalability of transformers in vision-related tasks [12]. Additionally, Touvron et al. presented Data-efficient Image Transformers (DeiT) in an effort to tackle the need for bigger amounts of data in transformers with specific training approaches that make learning efficient with small amounts of data [24]. Despite these developments, the application of transformers typically requires a higher amount of data and computations for achieving comparable results as CNNs. This is especially relevant in the area of agriculture where the availability of datasets is limited.

Additionally, the importance of transfer learning in previous research has also been discussed. In particular, Pan and Yang pointed out that transfer learning can help models pre-trained using large datasets to learn on smaller datasets more efficiently [15]. Such methods are widely applied in agriculture where access to large labeled datasets is difficult. The implementation of optimization algorithms like Adam was also suggested for accelerating model training [9].

In spite of numerous scientific studies in this field, a thorough comparative analysis that compares the performance of CNN-based and Transformer-based architectures under equal conditions still lacks. In other words, many studies compare different algorithms using one model or limited set of architectures which do not allow drawing any conclusions about performance, complexity, and generalization of these models. Also, previous research focuses on traditional convolutional neural networks or modern transformers independently, while an analysis of their advantages and disadvantages concerning particular problems is required.

It is important to emphasize that agricultural datasets can be characterized by certain peculiarities such as uneven lighting conditions, background noise, possible occlusion, and subtle visual difference between classes. Consequently, these characteristics influence the performance of any algorithm and require effective feature extraction techniques. As for the CNN architecture, it is well-known for its excellent feature extraction skills but, at the same time, Transformer-based architecture provides modeling the global context that can provide interesting results when applied to the particular problem.

Purpose of the Study

The goal of this study is to bridge the gap between previous studies by conducting an exhaustive comparative analysis of deep learning models such as convolutional neural networks (CNNs), transformers, and others in the field of fruit ripeness classification. The performance of each model will be measured based on metrics such as accuracy, precision, recall, and F1 score.

Table 1: Summary of CNN-Based Approaches in Related Work

Reference	**Model Used**	**Application**	**Key Finding**
Krizhevsky et al. [10]	AlexNet	Image classification	CNNs outperform traditional methods
He et al. [5]	ResNet	Image recognition	Residual learning improves deep networks
Huang et al. [7]	DenseNet	Image classification	Dense connections improve feature reuse
Kamilaris & Prenafeta-Boldú [8]	CNN-based models	Agriculture	Deep learning effective for agricultural tasks
Mohanty et al. [14]	CNN	Plant disease detection	High accuracy in agricultural classification
Aherwadi et al. [1]	CNN	Fruit maturity	Effective ripeness prediction

Table 2: Summary of Transformer-Based Approaches

Reference	**Model**	**Application**	**Key Finding**
Dosovitskiy et al. [3]	Vision Transformer	Image classification	Global feature modeling

Liu et al. [12]	Swin Transformer	Image recognition	Hierarchical transformer improves efficiency
Touvron et al. [24]	DeiT	Image classification	Data-efficient transformer training
Vaswani et al. [25]	Transformer	General	Self-attention enables global context

Table 3: Key Insights from Literature

Theme	**Insight**
CNN Strength	Strong local feature extraction and performance on moderate datasets
Transformer Strength	Captures global context but requires large datasets
Agricultural AI	Deep learning outperforms traditional methods
Research Gap	Limited comparative studies between CNN and transformer models

Methodology: In this methodology, there is a comprehensive study of various types of deep learning

methods to differentiate the fruits based on their ripeness. Methodology encompasses preparation,

pre-processing, modeling, training, and testing phases.

3.1 Data Preparation

The selected dataset that is being used in this research paper contains images of tomatoes, which have been collected from the FreshCheck Fruit Ripeness Database. All these images are divided into three categories of tomatoes based on their ripeness level. Specifically, three categories of tomatoes – unripe, ripe, and overripe tomatoes are used in this research. All these images included in this dataset were captured from the real world and depict various lighting and positioning scenarios of tomatoes. This data set has also been segregated into training and testing datasets in the ratio of 80:20

3.2 Data Preprocessing

For all the models considered in this case, the image dimensions are fixed to a size of 224×224 pixels. This dimension is commonly used in most deep learning platforms to make sure that there is compatibility with the pretrained network models. Normalization of the pixel intensities in a specific range increases the rate of convergence of the model.

The normalization equation is expressed as follows:

$x' = 255/x$

Rotation, flipping, and changing the brightness of the images is useful in overcoming overfitting.

3.3 Model Architectures

In this research work, a total of nine deep learning architectures have been examined. The deep learning architectures include CNN architecture and transformer architecture. Customized CNN architectures include customized CNN architecture, AlexNet, DenseNet201, and InceptionResNetV2. Transformers include ViT-B/32, DeiT Small, and Swin Transformer Tiny.

The CNN architectures are built using convolutional layers and pooling layers together with fully connected layers. Convolution operations follow the following formula:Customized CNN models involve convolution layers and pooling layers, as well as fully connected layers. Convolution is performed according to the following formula:

$$Y(i,j)=m\sum n\sum X(i+m,j+n)\cdot K(m,n) \text{ where}$$

Xrepresents the input feature map and $K$is the convolutional kernel.

Pooling layers are applied to reduce spatial dimensions while retaining essential features:

$$Y = \max_{f0}(X)$$

Fully connected layers transform the extracted features into class probabilities using the softmax function:

$$P(y = k) = \frac{ezk}{\sum_{j=1}^{K} e_z j}$$

where $z_k$represents the output score for class k.

DenseNet201 employs dense connectivity between layers, where each layer receives inputs from all preceding layers:

This improves gradient flow and encourages feature reuse.

Transformer-based models use self-attention mechanisms to model global dependencies across image patches. The self-attention mechanism is defined as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{=}\right)V$$

$$\sqrt{d_k}$$

where Q, K, and Vrepresent the query, key, and value matrices respectively.

3.4 Training Configuration

All models are trained using the categorical cross-entropy loss function:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y_i})$$

where $y_i$is the true label and $\hat{y_i}$is the predicted probability.

The Adam optimizer is used to update model parameters due to its adaptive learning rate capabilities:

$$\theta t+1 = \theta t - \alpha \frac{\hat{m_t}}{\sqrt{\hat{v_t}} + \epsilon}$$

where $\alpha$is the learning rate, and $\hat{m_t}$and $\hat{v_t}$represent the first and second moment estimates.

3.5 Evaluation Metrics

Model performance is evaluated using accuracy, precision, recall, and F1-score. Accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Recall is calculated as:

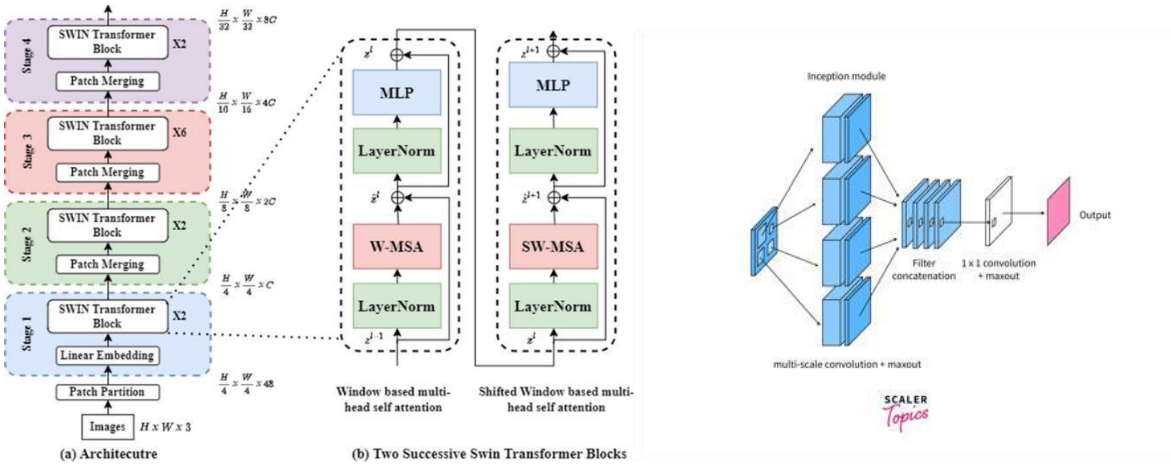
$$Recall = \frac{TP}{TP + FN}$$

The F1-score combines precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics provide a comprehensive evaluation of model performance across all classes.

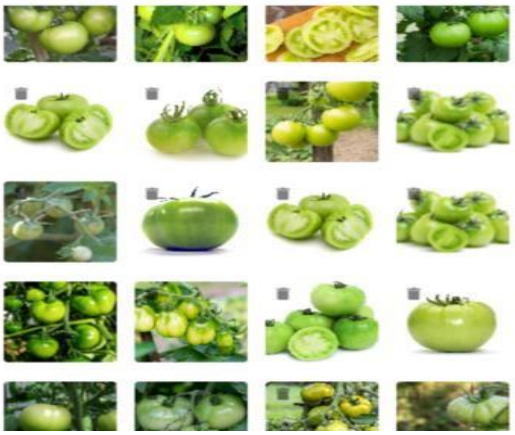
Architecture Diagram



Types of Ripeness-

Category: Tomato Maturity Detection				
Immature				
Original Image	Rotation	Zoom	Flipping	Scaling
				
Mature				
Original Image	Rotation	Zoom	Flipping	Scaling
				







Data Description:

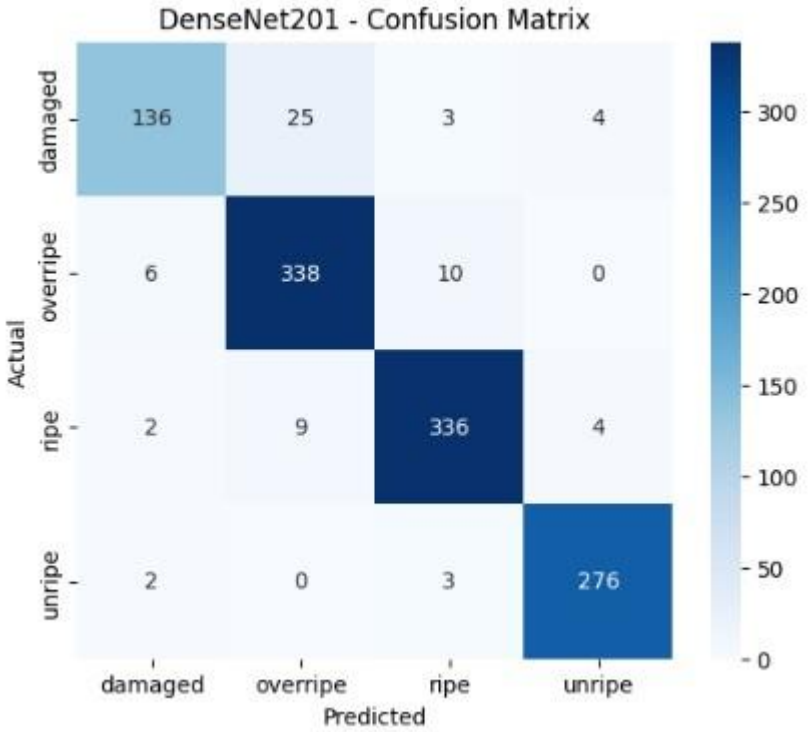
The dataset employed in the current research has been obtained from the FreshCheck Fruit Ripeness Classification dataset. However, for the current research purpose, only the tomato portion of the dataset has been used. The employed dataset includes real-world imagery collected through consumer-level cameras and mobile devices. Therefore, the variability in lighting, background, orientation of objects, and overall quality of images can be observed to make the task more realistic.

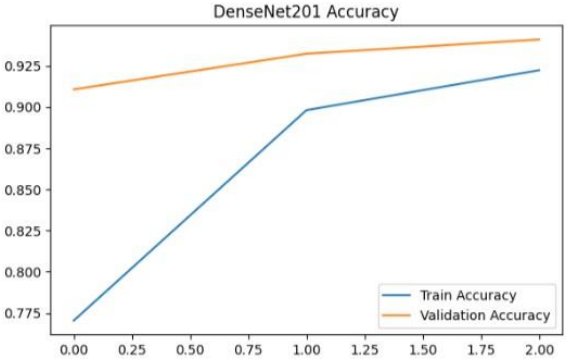
These classes consist of tomatoes with different degrees of ripeness, namely unripe, ripe, and overripe. The mentioned classes are appropriate for supervised image classification since they Each one of the states of maturity of the tomatoes corresponds to different states that can be distinguished by the naked eye. Such differences between the classes concern mainly the difference between the color, texture, and shape. However, in some cases, it is difficult to distinguish between the two classes because of their very similar appearance.

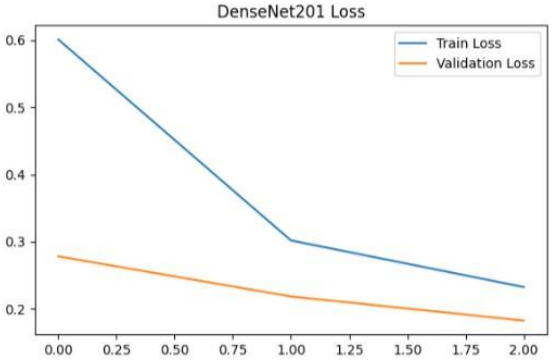
The images within the dataset had a dimension size that was normalized to be 224 x 224 pixels. This input dimension size corresponds to the input dimensions used by well-known deep learning architecture models such as CNNs and Transformers. Each pixel was normalized within all images in the dataset.

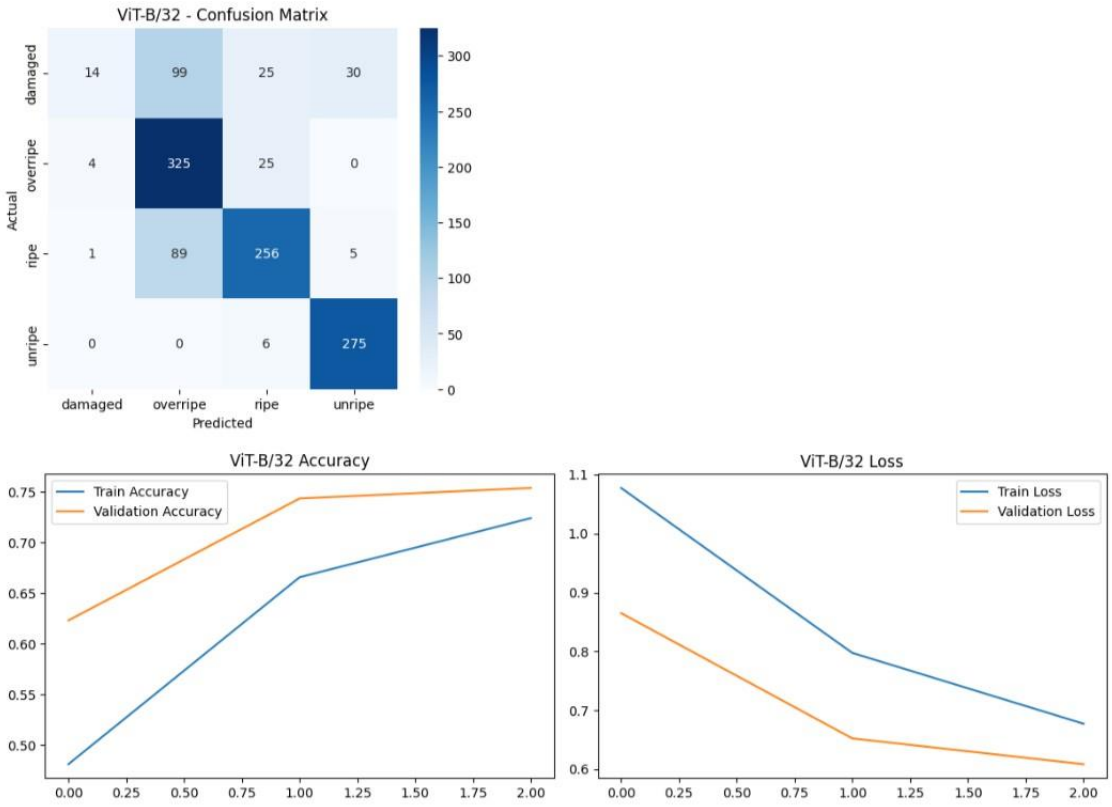
The division of data into training set data and validation set data was based on 80%-20% split ratio. The training set data would provide the source of data for training the model parameters, whereas the validation set would be useful for evaluating the performance of the model. The data is quite useful; however, it has several limitations.

Screenshots of test Results we got:


DenseNet201 - Confusion Matrix
damaged
overripe
ripe
unripe
Actual
136
25
3
4
6
338
10
0
2
9
336
4
2
0
3
276
300
250
200
150
100
50
0
damaged
overripe
ripe
unripe
Predicted


DenseNet201 Accuracy
0.925
0.900
0.875
0.850
0.825
0.800
0.775
Train Accuracy
Validation Accuracy
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00


DenseNet201 Loss
0.6
0.5
0.4
0.3
0.2
Train Loss
Validation Loss
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00



Results:

The performance of the proposed models was evaluated using certain classification evaluation metrics, which include accuracy, precision, recall, and F1 score. Nine models from the realm of deep learning were considered, including convolutional neural networks like AlexNet, DenseNet201, InceptionResNetV2, and three customized convolutional neural networks, along with transformer models like ViT-B/32, DeiT-Small, and Swin Transformer Tiny.

Table 1: Overall Model Performance

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
AlexNet	73.6%	78.6%	73.9%	68.7%
InceptionResNetV2	91.2%	91.2%	91.8%	91.6%

DenseNet201	**94.2%**	**94.8%**	**94.0%**	**94.9%**
ViT-B/32	75.1%	76.7%	75.9%	71.3%
DeiT-Small	73.5%	71.1%	73.3%	69.2%
Swin Transformer Tiny	71.6%	61.2%	71.4%	65.3%

DenseNet201 achieved the highest overall accuracy of 94.2%, indicating superior performance in feature extraction and classification. InceptionResNetV2 also performed strongly, achieving an accuracy above 91%. The custom CNN models demonstrated competitive results while using significantly fewer parameters.

Table 2: CNN vs Transformer Comparison

Category	**Best Model**	**Accuracy**
CNN Models	DenseNet201	94.2%
Transformer Models	ViT-B/32	75.1%

CNN-based models significantly outperformed transformer-based models. This difference can be attributed to the ability of CNNs to effectively learn local spatial features such as texture, shape, and color variation, which are critical for distinguishing fruit ripeness.

Table 3: Model Complexity vs Performance

Model	**Complexity**	**Performance**
AlexNet	Low	Moderate

InceptionResNetV2	High	High
DenseNet201	High	Very High
ViT-B/32	Very High	Moderate

Even though the transformer model is quite complex architecturally, it was still less efficient than the CNN architecture model. This may be contributed to by the fact that we had a small-sized dataset since the transformer model needs large-sized datasets for maximum efficiency.

Discussion:

From the results of the experiment, it is clear that the CNN architecture model is more efficient in solving our problem than the transformer model. This is because the DenseNet201 was very accurate because of the dense architecture of the model.

InceptionResNetV2 performed very well because of the inclusion of inception layers and residual connections, which help in capturing features at different scales. Therefore, it would be wise to use CNN models as they do a great job in identifying even slight changes in features such as texture or color change. However, transformer models were not successful, and there are many reasons behind this failure. For example, while transformer models have some strengths that enable them to capture relationships in images better, they are highly data-hungry models. Hence, the performance of these models in small datasets is insignificant. Moreover, there are practical issues as well including variable lighting conditions, noise, and different angles of fruits.

While CNNs showed higher resistance against such problems, the same was not the case with transformer models, which required a more extensive dataset.

It is evident from these results that when choosing an appropriate model, one must take into consideration the nature of the dataset. With a small number of samples in the dataset, CNNs performed better than transformers.

Conclusion: In this study, we have performed a comparative study on different deep learning architectures for classifying the ripeness stage of fruits. We have used nine different architectures for classification, which include both CNN architectures and transformer architectures. Our experiment has been done using a real-world tomato dataset having three classes such as unripe, ripe, and overripe tomatoes. We have found that CNN architectures perform much better than transformer architectures for the problem statement mentioned above. The DenseNet201 architecture achieves the maximum accuracy of 94.2%, whereas the second-best accuracy is achieved by the InceptionResNetV2 architecture having an accuracy of 91.2%, which proves that deep CNN architectures are more suitable for extracting discriminative features from images. In case of transformer architectures, poor results have been observed, which might be attributed to the smallness of the dataset size.

We conclude that CNN architectures are still useful for classification problems based on image datasets.

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